

ORIGINAL CONTRIBUTION

Updated Meta-Analysis of Left Atrial Appendage Closure Versus Oral Anticoagulation to Prevent Stroke in Atrial Fibrillation

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BACKGROUND: Ninety percent of left atrial thrombi in nonvalvular atrial fibrillation arise in the left atrial appendage, which can be occluded percutaneously (left atrial appendage closure [LAAC]). Two 2026 trials (CHAMPION-AF and CLOSURE-AF) more than doubled the available randomized evidence comparing LAAC with oral anticoagulation. We evaluated pooled effects of LAAC on stroke, bleeding, and mortality by bleeding risk, anticoagulation eligibility, and comparator.

METHODS: In this systematic review and meta-analysis, we searched PubMed, Scopus, and Embase from inception to April 15, 2026, for randomized controlled trials comparing percutaneous LAAC with oral anticoagulation in nonvalvular atrial fibrillation. Risk of bias was assessed with the Cochrane Risk of Bias 2 tool. Cumulative-incidence outcomes were pooled as odds ratios using random-effects models, with prediction intervals, prespecified subgroups, and trial sequential analysis.

RESULTS: Six randomized controlled trials (2014–2026; 358 sites in North America, Europe, and Asia; 7028 patients) compared LAAC with warfarin (2 trials) or a direct oral anticoagulant (4 trials); populations were predominantly anticoagulation-eligible with low prior-stroke prevalence, plus 2 high-bleeding-risk and 1 postablation trial. Overall stroke rates were similar between arms (odds ratio, 1.10 [95% CI, 0.78–1.56]). Ischemic stroke was more frequent with LAAC (odds ratio, 1.41 [95% CI, 1.07–1.86]) but did not reach the required information size. Nonprocedural bleeding was lower with LAAC (odds ratio, 0.58 [95% CI, 0.35–0.96]), the only outcome reaching a sufficient information size, in standard-risk and postablation but not high-bleeding-risk patients.

CONCLUSIONS: LAAC is best positioned to reduce nonprocedural bleeding in anticoagulation-eligible patients at standard-to-moderate bleeding risk, rather than as a stroke-equivalent substitute for oral anticoagulation.

REGISTRATION: URL: <https://www.crd.york.ac.uk/PROSPERO/>; Unique identifier: CRD420261370546.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: atrial fibrillation ■ hemorrhagic risk ■ ischemic stroke ■ left atrial appendage ■ oral anticoagulation

Atrial fibrillation (AF) affects an estimated 37.5 million people worldwide, and its global prevalence is projected to increase by >60% by 2050, driven by population aging and the growing burden of cardiometabolic

risk factors.¹ Cardioembolic events originating from the heart are associated with ≈20% to 25% of all ischemic strokes.² Among these, AF is the most common cardiac source of embolism, associated with increasing the risk

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This manuscript was sent to Harold P. Adams, Jr., Guest Editor, for review by expert referees, editorial decision, and final disposition.

Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/STROKEAHA.126.057089>.

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Stroke is available at www.ahajournals.org/journal/str

Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
DOAC	direct oral anticoagulant
CHA₂DS₂-VASC	congestive heart failure, hypertension, age, diabetes, stroke, vascular disease, age, sex score
HAS-BLED	hypertension, abnormal renal/liver function, stroke, bleeding history or predisposition, labile international normalized ratio, elderly, drugs/alcohol concomitantly score
ISTH	International Society on Thrombosis and Haemostasis
LAA	left atrial appendage
LAAC	left atrial appendage closure
LFK	Luis Furuya-Kanamori index
OAC	oral anticoagulation
OR	odds ratio
RCT	randomized controlled trial
RIS	required information size
TSA	trial sequential analysis

of ischemic stroke by 3- to 5-fold and contributing to an estimated 15% of all strokes worldwide.²

Oral anticoagulation (OAC), first with vitamin K antagonists and later with direct oral anticoagulants (DOACs), has transformed outcomes in patients with AF at high thromboembolic risk. DOACs offer more predictable pharmacokinetics, fewer drug interactions, and lower rates of intracranial hemorrhage than vitamin K antagonists and are now recommended for most patients with clinically significant stroke risk.³ Nevertheless, the need for lifelong anticoagulation must be balanced against modifiable bleeding risk (driven potentially by concomitant antiplatelet or nonsteroidal anti-inflammatory drug use), nonmodifiable factors such as chronic renal dysfunction and frailty-related falls, and the practical limitations of long-term therapy (adherence, drug interactions, and patient preference).³

These limitations have created interest in nonpharmacologic stroke prevention. Because >90% of left atrial thrombi in patients with nonvalvular AF originate within the left atrial appendage (LAA), percutaneous LAA closure (LAAC) was developed to mechanically occlude this embolic source and deliver indefinite systemic anticoagulation.⁴ Since US regulatory approval was granted in 2015, the Watchman/Watchman FLX platform has demonstrated procedural success exceeding 98% at high-volume centers, with pericardial effusion rates of 1% to 2% and device-related thrombus occurring in ≈3% to 4% of implants.⁵ Broader improvements in device design, implantation technique, periprocedural imaging, and operator experience have further

enhanced safety and accelerated adoption across many healthcare systems.⁶

A 2025 pooled analysis by Kaisaier et al⁷ that incorporated available randomized controlled trials (RCTs) comparing LAAC and anticoagulation in patients who could not tolerate or were contraindicated to OAC suggested that LAAC may be associated with lower rates of hemorrhagic stroke and nonprocedural bleeding, as well as a potentially lower mortality rate, without an apparent excess in overall thromboembolic events. However, the cumulative sample size was modest, leaving the study underpowered for several clinically relevant end points and pooled estimates potentially sensitive to the addition of any single large trial.

The recent publication of 2 major RCTs, CHAMPION-AF⁸ and CLOSURE-AF,⁹ materially changed this landscape by more than doubling the previously available randomized evidence base. The CLOSURE-AF trial enrolled patients with AF at high stroke and bleeding risk and compared LAAC against physician-directed best medical therapy, including anticoagulation when appropriate. CLOSURE-AF demonstrated event rates that numerically favored medical therapy over LAAC, a cautionary signal suggesting that device-based strategies may not uniformly outperform optimized contemporary care in medically complex populations.⁹ In contrast, CHAMPION-AF demonstrated noninferiority for major efficacy outcomes and superiority for nonprocedure-related bleeding of LAAC compared with contemporary DOAC therapy in selected patients eligible for long-term anticoagulation.⁸

Given the heterogeneity in the populations examined to date, an updated systematic review and meta-analysis is necessary. In this evaluation, we have included all RCTs comparing LAAC with OAC. We aimed to generate robust pooled estimates for stroke, bleeding, and mortality; to determine whether prior favorable signals are sustained or attenuated with the expanded evidence base; and to explore treatment-effect heterogeneity by bleeding risk, anticoagulation eligibility, and comparator regimen.

METHODS

Data Availability

All analyzed data are from previously published, publicly available trials; no new patient-level data were generated. All data are publicly available within the article, [Online Supplemental File](#), or publicly available sources. No other information is needed to reproduce these results.

Study Design

We used the PRISMA 2020 guidelines¹⁰ to prepare this systematic review and meta-analysis. The protocol was prospectively registered in PROSPERO (REGISTRATION: URL: <https://www.crd.york.ac.uk/PROSPERO/>; Unique identifier: CRD420261370546).

Search Strategy and Study Selection

We searched PubMed, Scopus, and Embase from database inception through April 15, 2026, for RCTs that compared percutaneous LAAC with OAC for stroke prevention in patients with nonvalvular AF. The full search strategy for each database is shown in [Table S1](#). To identify any additional eligible trials, we also performed forward citation tracking and backward reference searching of included studies and relevant systematic reviews using Citationchaser.¹¹ Screening and selection were managed in Covidence (Veritas Health Innovation, Melbourne, Australia). When a single trial had multiple publications with different follow-up intervals, we took data from the publication with the longest available follow-up. We did not perform an individual-patient-data meta-analysis; for PROTECT AF and PREVAIL, aggregate event counts came from the published pooled analysis by Reddy et al¹² This was not a prespecified aim and affected only the pooled PROTECT AF/PREVAIL warfarin unit.

Data Extraction

One reviewer (A.M.) performed study selection, data extraction, and risk-of-bias assessment in the Covidence platform using a preformatted Excel extraction template; a second reviewer (P.K.P.B.) independently cross-verified all extracted data against the original publications. Disagreements were resolved by consensus, and a third reviewer (S.K.F.) was consulted where contention remained. Extracted variables covered study design, patient demographics, baseline characteristics (including congestive heart failure, hypertension, age, diabetes, stroke, vascular disease, age, sex score [CHA₂DS₂-VASc] and hypertension, abnormal renal/liver function, stroke, bleeding history or predisposition, labile international normalized ratio, elderly, drugs/alcohol concomitantly score [HAS-BLED] scores), intervention and comparator details, and every prespecified efficacy and safety outcome. For each outcome, we recorded the number of patients with at least 1 event and the total number of patients randomized per treatment arm.

Outcomes

The primary outcomes were all stroke (ischemic and hemorrhagic, excluding transient ischemic attack), ischemic stroke, and hemorrhagic stroke. Secondary outcomes were all-cause death and nonprocedural clinically relevant bleeding (International Society on Thrombosis and Haemostasis [ISTH] major plus clinically relevant nonmajor bleeding). Subgroup variables were comparator class (vitamin K antagonist versus DOAC); stroke risk profile (mean CHA₂DS₂-VASc score ≤ 3.5 versus > 3.5); prevalence of prior stroke or transient ischemic attack ($< 20\%$ versus $\geq 20\%$); HAS-BLED category (mean < 3 versus ≥ 3); bleeding risk at trial enrollment, defined by inclusion criteria as standard risk (trials enrolling anticoagulation-eligible patients), high risk (trials enrolling patients with a documented or relative contraindication to OAC), or postablation (trials enrolling patients undergoing concurrent AF ablation); and device generation (first-generation Watchman versus Watchman FLX versus contemporary or mixed devices). When end point definitions differed across trials for a given outcome, trials were pooled only where the definitions were clinically comparable; any exclusions or remaining definition heterogeneity are

flagged alongside the corresponding forest plot. Outcomes were the cumulative incidence of the first event (patients with ≥ 1 event per number randomized), pooled as odds ratios (ORs); the longest available follow-up was used.

Risk of Bias Assessment

Risk of bias was assessed separately for efficacy and safety end points with the Cochrane Risk of Bias 2 tool.¹³ Assessments were anchored to the original trial publication because the tool evaluates trial conduct (randomization, blinding, and outcome ascertainment) established at trial inception. For domains potentially affected by extended follow-up (missing outcome data and selective reporting), judgments were reevaluated against the follow-up publication. Results are shown as traffic light plots and summary bar plots. Domain-level rationale is given in full in [Table S2](#) and [Figures S1](#) and [S2](#). Risk-of-bias judgments (Cochrane Risk of Bias 2¹³) informed the certainty of evidence but did not weight or exclude trials. Risk-of-bias assessment for included trials is available in [Table S3](#).

Statistical Analysis

All analyses were done in RStudio with R, version 4.5.2 (Vienna, Austria). For every dichotomous outcome, pooled ORs with 95% CIs were calculated using the metabin function. Both random-effects and fixed-effect models were generated for each outcome; the random-effects model with the restricted maximum likelihood estimator served as the preferred primary analysis. For comparisons based on only 2 trials, both the random-effects and fixed-effect estimates are reported in the main text. Between-study heterogeneity was quantified with the I^2 statistic and the Cochran Q test. For every pooled estimate, a prediction interval was also calculated to convey the range of treatment effects expected in similar future settings. The random-effects model (restricted maximum likelihood with the Hartung-Knapp adjustment) was primary. For the 2-trial warfarin comparisons, the random-effects interval is uninformatively wide, so the fixed-effect estimate is reported in the text (all estimates in the [Supplemental Material](#)). The prediction interval¹⁴ reflects heterogeneity and the range expected in a future study (reported only with ≥ 3 trials), unlike the CI for the pooled mean. A prediction interval much wider than the CI, or 1 that crosses the null when the CI does not, indicates that the average effect may not hold in every setting.

A leave-one-out sensitivity analysis was performed for each outcome, sequentially dropping every study to gauge the influence of individual trials on the pooled estimate. Small-study effects and publication bias were assessed using the Doi plot with the Luis Furuya-Kanamori (LFK) index, classified as no asymmetry (LFK -1 to $+1$), minor asymmetry (-2 to -1 or $+1$ to $+2$), or major asymmetry (< -2 or $> +2$).

Prespecified subgroup analyses were performed for all outcomes using study-level characteristics. Interactions between subgroups were tested with the χ^2 test for heterogeneity under both the common-effect and random-effects models; where these produced divergent inferences, both results are reported. The subgroup variables are defined above. Where stratification left a single trial in a subgroup, the trial-level estimate is reported without pooling. Deviations from the PROSPERO-registered protocol were as follows. First, primary efficacy was

reported as 3 separate stroke end points (all stroke, ischemic, hemorrhagic) rather than the registered composite of stroke, systemic embolism, and cardiovascular death, because trial-level composites used different components and could not be pooled meaningfully. Second, primary safety was operationalized as nonprocedural ISTH major plus clinically relevant non-major bleeding rather than trial-defined major bleeding. Third, the HAS-BLED cutoff was redefined from 2 to 3 to match how the score was reported in the included trials. Fourth, the postablation subgroup was expanded into a 3-level bleeding risk-at-enrollment stratification. Fifth, baseline stroke or transient ischemic attack prevalence and device generation were added as stratifications. Sixth, the trial-result-direction stratification was dropped because only 1 trial failed noninferiority assessment. Outcomes registered but not analyzed here (systemic embolism, cardiovascular death, device-related thrombus, pericardial effusion, and net clinical benefit) had sparse events or definitional heterogeneity that precluded pooling. The subgroup analysis (vitamin K antagonist versus DOAC) corresponds to the PROSPERO-registered sensitivity analysis isolating the LAAC versus DOAC comparison. Subgroup interaction was assessed with the *Q*-based test for subgroup differences¹⁵ (random-effects primary; common-effect as sensitivity).

Trial Sequential Analysis

Trial sequential analysis (TSA) was performed using TSA software (v0.9.5.10 Beta, Copenhagen Trial Unit, Denmark) for each outcome to assess whether the cumulative evidence was sufficient to support firm conclusions. TSA controls type I error inflation from repeatedly updating a cumulative meta-analysis and estimates the required information size (RIS), a meta-analytic sample size^{16,17}; evidence was considered firm only when the cumulative *Z*-curve crossed the monitoring boundary or the RIS was reached.

RIS was calculated with a 2-sided α of 0.05, 80% power, the pooled control-arm event rate, and a prespecified minimum clinically relevant effect on the OR scale. Between-trial heterogeneity was built in via the *D*² diversity estimator, and O'Brien-Fleming monitoring boundaries were applied to the cumulative *Z*-score. Outcomes for which the monitoring boundary was not crossed and the RIS was not reached were classed as inconclusive. TSA results are presented alongside conventional estimates to inform the certainty of evidence (Supplemental Methods and Table S3).

Reporting Guidelines

This systematic review and meta-analysis were conducted and reported in accordance with the PRISMA 2020 guidelines.¹⁰ A complete PRISMA checklist is provided in Table S4.

RESULTS

Systematic Search

The database search returned 304 records (PubMed *n*=49, Embase *n*=95, Scopus *n*=160), which fell to 278 after the removal of 26 duplicates (Figure). Title and abstract screening eliminated 268 records, leaving 10 reports for full-text review. Two reports were

excluded for wrong comparator arms. Of the final 8 reports, 4 were publications of trials already included in prior meta-analyses and added no new data, 2 were extensions of prior RCTs providing long-term follow-up data, and 2 were new RCTs that met inclusion criteria (CHAMPION-AF and CLOSURE-AF). Forward citation tracking (2365 records) and backward reference searching (115 records) flagged another 4 reports for full-text assessment, 3 of which were excluded because they were the wrong study type and 1 because it used the wrong comparator. The final analysis included 6 unique RCTs reported across these 8 publications: 4 trials were carried forward from prior meta-analyses (PROTECT AF, PREVAIL, PRAGUE-17, and OPTION),¹⁸⁻²¹ with longer follow-up available for 3 of them through 2 extension papers (Reddy et al¹² reported 5-year outcomes for PROTECT AF and PREVAIL combined, and Osmancik et al²² reported 4-year outcomes for PRAGUE-17). Event counts were extracted from the longest available follow-up for each trial. The 2 new RCTs were CHAMPION-AF and CLOSURE-AF.

Risk of bias for the efficacy end point was rated Low in CHAMPION-AF and OPTION and Some Concerns in CLOSURE-AF, PRAGUE-17, PROTECT AF, and PREVAIL, driven primarily by open-label designs and partially unblinded outcome adjudication. Risk of bias for safety end points was Some Concerns across all trials, reflecting subjectivity in bleeding ascertainment.

Study and Population Characteristics

The 6 included RCTs randomized 7028 patients with non-valvular AF to LAAC or OAC across 358 sites in North America, Europe, and Asia. Trial-level characteristics are shown in full in Table 1, and comorbidity characteristics are described in Table 2.

The mean age across the RCTs ranged from 69.6 years to 77.9 years, and the proportion of women ranged from 29.9% to 38.6%. Baseline stroke risk spanned a wide gradient: mean CHA₂DS₂-VASc was 3.5 in CHAMPION-AF, OPTION, and PROTECT AF, 4.0 in PREVAIL, 4.7 in PRAGUE-17, and 5.2 in CLOSURE-AF (Table 1). PROTECT AF and PREVAIL used warfarin as the comparator paired with first-generation Watchman, whereas the 4 contemporary trials used a DOAC paired with Watchman FLX, Amulet, or a mix of devices. In PRAGUE-17, 95.5% of patients in the DOAC arm received apixaban. Trial populations also differed by bleeding risk at enrollment: CHAMPION-AF, PROTECT AF, and PREVAIL enrolled patients suitable for long-term OAC; PRAGUE-17 and CLOSURE-AF enrolled patients at high bleeding risk on OAC; and OPTION enrolled exclusively postablation patients. Follow-up was 3 years in the 3 most recent trials and 5 years in PROTECT AF and PREVAIL, with a median of 3.5 years in the PRAGUE-17 4-year report.

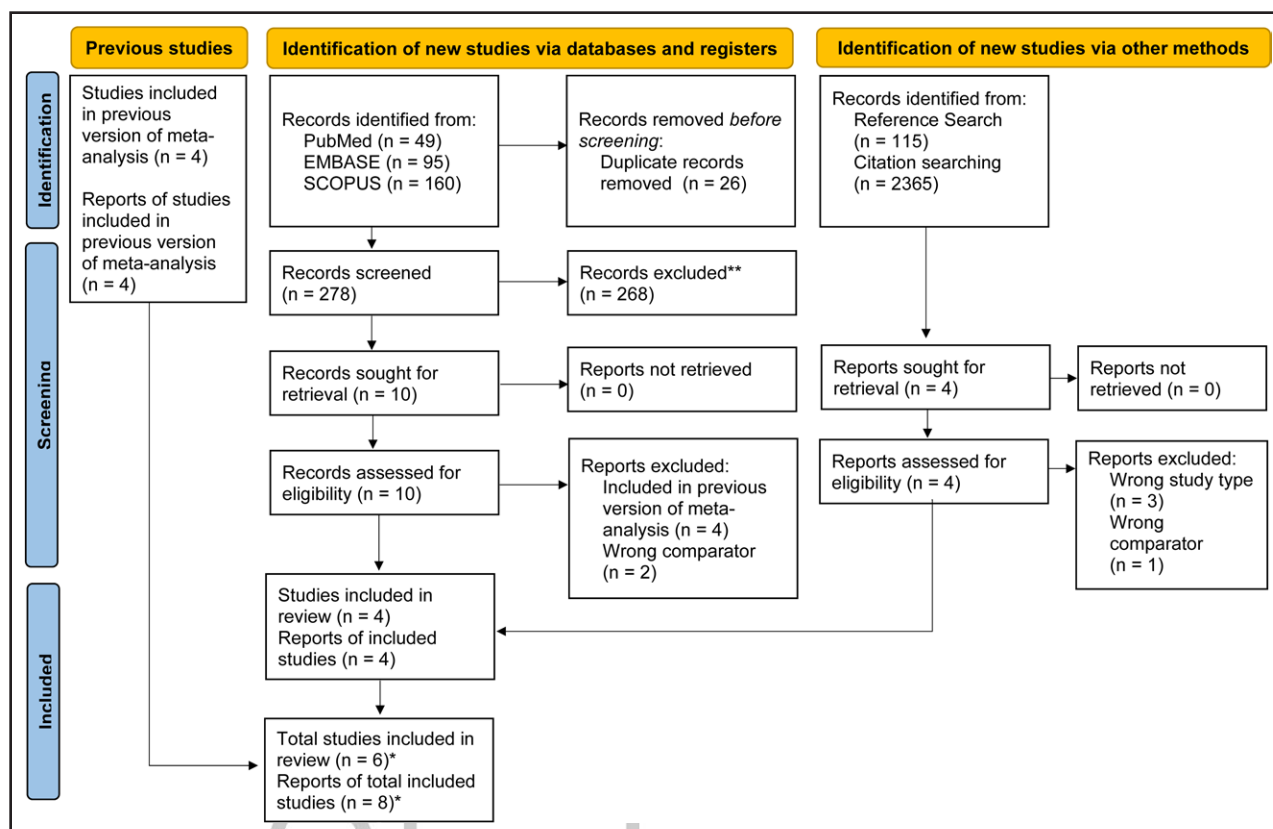


Figure. PRISMA flowchart of systematic search.

*PRAGUE-17 includes 2- and 4-year follow-up reports; IPD meta-analysis of PROTECT-AF and PREVAIL included as report. †Exclusions by human reviewers during title and abstract screening.

Primary Pooled Analyses

Pooled estimates, prediction intervals, and robustness and TSA findings are summarized in Table 3. Forest plots, leave-one-out plots, Doi/LFK plots, and TSA plots are presented in Figures S3 through S22. All pooled estimates use a random-effects model with Hartung-Knapp adjustment.

All Stroke Subgroup

Overall stroke rates were similar between arms (6 trials; OR, 1.10 [95% CI, 0.78–1.56]; $P=10.6\%$; prediction interval, 0.60–2.02); TSA accrued only 29% of the RIS, so this comparison was inconclusive (Table 3). When differentiating stroke by type, the direction diverges. Ischemic stroke was more frequent with LAAC than with OAC (OR, 1.41 [95% CI, 1.07–1.86]; $P=0\%$; prediction interval, 0.96–2.08), and every leave-one-out iteration continued to favor OAC (OR range, 1.27–1.49). The Doi plot showed no asymmetry (LFK, 0.24). The cumulative Z-score crossed the conventional 1.96 threshold but did not cross the TSA α -spending boundary, with only 36% of the RIS accrued, meaning the ischemic stroke signal did not reach the threshold for firm inference. Hemorrhagic stroke did not differ between groups (OR, 0.55 [95% CI, 0.25–1.22]; $P=16.4\%$; prediction interval 0.14–2.11). The point

estimate here was driven by PROTECT AF; dropping that trial shifted the point estimate to OR 0.75 (95% CI, 0.48–1.17) without changing the inference of no significant difference between groups, and TSA stayed below the α -spending boundary at 63% of the RIS. The all-stroke risk did not differ between LAAC and DOACs (OR, 1.23 [95% CI, 0.82–1.84]) or between LAAC and warfarin (fixed-effect OR, 0.86 [95% CI, 0.53–1.41]; random-effects OR, 0.90 [95% CI, 0.01–101.91]).

Ischemic Stroke Subgroup

LAAC was associated with a higher rate of ischemic stroke in every subgroup examined. Point estimates favored OAC in both the standard-bleeding-risk stratum (OR, 1.63 [95% CI, 0.97–2.72]) and the high-bleeding-risk stratum (OR, 1.25 [95% CI, 0.68–2.30]); in DOAC-controlled trials (OR, 1.38 [95% CI, 0.91–2.08]) and warfarin-controlled trials (fixed-effect OR, 1.56 [95% CI, 0.84–2.90]; random-effects OR, 1.53 [95% CI, 0.05–44.21]); and in every device generation, HAS-BLED, and CHA₂DS₂-VASc subgroup. No interaction tests were significant ($P>0.40$), and within-subgroup heterogeneity was zero ($P=0\%$). OPTION, which exclusively enrolled postablation patients, was the only trial in which the point estimate did not favor OAC (OR, 0.89 [95% CI, 0.36–2.21]). None of the

Table 1. Overview of Included Studies

Trial	Country; sites	No. of patients (LAAC/OAC)	Population age (y); % female; CHA ₂ DS ₂ -VASc score	LAAC and comparator	Follow-up length	Primary efficacy: end point and result	Primary safety end point and result
CHAMPION-AF ⁸	16 countries; 141 sites	3000 (1499/1501)	71.7±7.5; 31.9%; 3.5±1.3	Watchman FLX vs DOAC (physician choice)	3 y	Composite: CV death, stroke, SE	Nonprocedural bleeding (ISTH major + CRNMB)
CLOSURE-AF ⁹	Germany; 42 sites	912 (446/442)	77.9±7.1; 38.6%; 5.2±1.5	Watchman/Watchman FLX or Amulet vs DOAC (physician choice)	Median 3.0 y	Composite: stroke, SE, major bleeding (BARC ≥3), CV/unexplained death	Included in primary composite Major bleeding (BARC ≥3)
OPTION ²¹	10 countries; 106 sites	1600 (803/797)	69.6±7.7*; 34.1%*; 3.5±1.3*	Watchman FLX (concomitant/sequential with AF ablation) vs DOAC (physician choice)	36 mo	Composite: death (any cause), stroke, SE	Nonprocedural bleeding (ISTH major + CRNMB)
PRAGUE-17 ^{20,22}	Czech Republic; 10 sites	402 (mITT) (201/201)	73.3; 34.3%; 4.7±1.5	Watchman/Amulet vs DOAC (apixaban 95.5%)	Median 3.5 y (4-y report)	Composite: stroke/TIA, SE, CV death, clinically significant bleeding, procedural complications	Included in primary composite Clinically relevant bleeding (ISTH major + CRNMB)
PROTECT AF ¹⁸	USA, Europe; 59 sites	707 (463/244)	72.0±8.9; 29.9%; 3.5±1.6	Watchman (1st gen) vs warfarin (INR 2–3)	5 y (mean 47.6 mo; 2717 PY)	Composite: stroke, SE, CV/unexplained death	Procedural events + major bleeding
PREVAIL ¹⁹	USA; ≤50 sites	407 (269/138)	74.0±7.4; 29.9%; 4.0±1.2	Watchman (first gen) vs warfarin (INR 2–3)	5 y (mean 47.9 mo; 1626 PY)	First coprimary: stroke, SE, CV/unexplained death Second coprimary: ischemic stroke/SE >7 d	Early safety (≤7 d): death, stroke, SE, or device/procedure events
PROTECT AF + PREVAIL ^{12†}	USA, Europe; 59+50 sites	1114 (732/382)	72.6±8.4/73.5±8.6; 29.9%; 3.6/3.9	Watchman (first gen) vs warfarin (INR 2–3)	5 y (4343 PY)	Patient-level meta-analysis: Stroke/SE/CV death, all stroke/SE, hemorrhagic stroke, all-cause death	Major bleeding (all): HR, 0.91 (0.64–1.29), <i>P</i> =0.60 Nonprocedural major bleed: HR, 0.48 (0.32–0.71), <i>P</i> =0.0003

AF indicates atrial fibrillation; BARC, Bleeding Academic Research Consortium score; CHA₂DS₂-VASc, congestive heart failure, hypertension, age, diabetes, stroke, vascular disease, age, sex score; CRNMB, clinically relevant nonmajor bleeding; CV, cardiovascular; DOAC, direct oral anticoagulant; HR, hazard ratio; INR, international normalized ratio; ISTH, International Society on Thrombosis and Haemostasis; LAAC, left atrial appendage closure; mITT, modified intention-to-treat; NMCBR, nonmajor clinically relevant bleeding; OAC, oral anticoagulation; PY, patient-years; SE, systemic embolism; and TIA, transient ischemic attack.

*Population after ablation.

†Individual patient-level meta-analysis.

subgroup-specific estimates reached statistical significance individually; the consistent direction across strata is descriptive rather than inferential.

Hemorrhagic Stroke Subgroup

LAAC reduced the risk of hemorrhagic stroke compared with warfarin (fixed-effect OR, 0.20 [95% CI, 0.07–0.55]; random-effects OR, 0.20 [95% CI, 0.00–23.79]) but not DOACs (OR, 0.82 [95% CI, 0.58–1.16]; Table 4). The difference between these 2 comparator subgroups was significant (*P*=0.0003). A similar but weaker pattern was seen for all-cause mortality: LAAC was associated with lower mortality than warfarin (OR, 0.72 [95% CI, 0.52–0.99]) but not DOACs (OR, 1.00 [95% CI, 0.74–1.35]; *P*=0.08). First-generation Watchman was used only in the warfarin-controlled trials (OR, 0.20), whereas Watchman FLX and contemporary or mixed-device trials were DOAC controlled (OR, 1.00 and OR, 0.72, respectively).

All-Cause Mortality Subgroup

All-cause mortality did not differ between LAAC and OAC (*k*=5, with PROTECT AF and PREVAIL pooled; OR, 0.91 [95% CI, 0.69–1.20]; *P*=34.7%; 95% prediction interval, 0.55–1.51; Table 3). The estimate was carried by the long-term PROTECT AF and PREVAIL follow-up data; removing those 2 trials pooled together shifted the OR to 1.00 (95% CI, 0.74–1.35) without changing the inference of no significant mortality difference. The Doi plot showed major small-study asymmetry (LFK, −2.59). TSA reached 67% of the RIS without crossing the α -spending boundary, meaning the mortality comparison is inconclusive.

Bleeding Subgroups

Nonprocedural clinically relevant bleeding was lower with LAAC than with OAC (*k*=5, with PROTECT AF and PREVAIL pooled; OR, 0.58 [95% CI, 0.35–0.96]; *P*=78.8%; prediction interval, 0.19–1.80; Table 3). The

Table 2. Baseline Comorbidity Characteristics Across the Included Randomized Trials of Left Atrial Appendage Closure Versus Oral Anticoagulation

Trial	% White*	BMI (mean ± SD)	AF type† (paroxysmal; persistent; permanent)	HAS-BLED score‡ (mean ± SD)	LVEF§ (mean ± SD)	Hypertension (%)	Diabetes (%)	HF/CMP (%)	CAD (%)	Prior stroke or TIA (%)	Prior major bleeding¶ (%)
CHAMPION-AF#	85.1	NR	68.9%; 24.7%; 6.4%	1.3±0.8	NR	NR	NR	NR	NR	8.2	NR
CLOSURE-AF**††	93.6	NR	NR	3.0±0.9	NR	93.9	40.7	17.4	55.6	33.0	30.4
OPTION‡‡	84.9	30.7±6.0	61.1%; 38.9%; –	1.2±0.8	NR	88.9	27.7	20.4	31.4	10.7	NR
PRAGUE-17‡‡,‡‡	NR	NR	29.9%; 23.1% (long-standing persistent 8.5%); 38.6%	3.0±0.9	53.1±12.4	92.5	40.5	44.3	NR	32.1	50.7
PROTECT AF‡‡	91.5	31.5±6.1	42.3%; 20.8%; 35.8%	NR	57.1±9.8	89.8	26.2	26.9	NR	18.5	NR
PREVAIL‡‡	94.3	NR	49.6%; 30.5%; 15.7%	NR	55.6±9.9	91.4	32.4	23.3	NR	27.8	NR
PROTECT AF + PREVAIL‡‡‡	92.5	31.5±6.1	45.0%; 24.3%; 28.5%	NR	56.6±9.8	90.4	28.5	25.6	NR	21.9	NR

Values are pooled means or proportions across treatment arms within each trial, weighted by arm sample size. Continuous variables are presented as mean ± SD; categorical variables as percentages. AF indicates atrial fibrillation; BARC, Bleeding Academic Research Consortium; BMI, body mass index; CAD, coronary artery disease; CMP, cardiomyopathy; HAS-BLED, hypertension, abnormal renal/liver function, stroke, bleeding history, labile International Normalized Ratio, elderly, drugs/alcohol; HF, heart failure; LVEF, left ventricular ejection fraction; NR, not reported; TIA, transient ischemic attack; and –, category not enrolled.

*Race was self-reported. The OPTION and CHAMPION-AF trials had a substantial proportion of participants (10% to 12%) for whom race or ethnicity was not disclosed; the proportions shown reflect classified responses.

†AF pattern was not reported in the CLOSURE-AF baseline characteristics. The OPTION cohort only enrolled patients undergoing or recently having undergone catheter ablation; patients with permanent AF were not enrolled. Long-standing persistent AF was reported as a discrete category only in PRAGUE-17.

‡PROTECT AF, PREVAIL, and the 2017 pooled analysis did not report HAS-BLED scores at baseline.

§Baseline LVEF was reported only by the 3 earliest trials. OPTION, CHAMPION-AF, and CLOSURE-AF each excluded patients with LVEF <30% per protocol but did not tabulate baseline values.

¶Prior major bleeding was defined as BARC type 3a or 3b in CLOSURE-AF (an additional 12.3% of the cohort had BARC 3c intracranial bleeding, a qualifying inclusion criterion) and as history of bleeding or bleeding predisposition in PRAGUE-17. Recent major bleeding was an exclusion criterion in PROTECT AF, PREVAIL, OPTION, and CHAMPION-AF.

#Combined ischemic and hemorrhagic stroke for CHAMPION-AF (reported separately as 7.8% and 0.5%, respectively).

**CLOSURE-AF reported the comorbidity as cardiomyopathy rather than congestive heart failure; the 2 are aggregated for cross-trial comparison.

††CLOSURE-AF reported coronary heart disease, inclusive of prior myocardial infarction.

‡‡The PROTECT AF + PREVAIL row reflects sample size-weighted averages across the 2 individual patient-level meta-analysis source trials (n=1114) and is included for direct row-correspondence with Table 1.

heterogeneity was driven by CLOSURE-AF, which favored OAC; removing that trial yielded a tight pooled benefit with no residual heterogeneity (OR, 0.50 [95% CI, 0.40–0.61]; $P=0\%$). TSA crossed the α -spending boundary at 78% of the RIS, and the Doi plot was symmetrical (LFK, 0.12). This was the only outcome for which the cumulative randomized evidence reached a statistically validated conclusion.

It is important to note that the benefit of LAAC depended on a patient's baseline bleeding risk when on anticoagulation. Among trials that enrolled patients at standard bleeding risk, LAAC reduced nonprocedural clinically relevant bleeding (OR, 0.53 [95% CI, 0.44–0.64]; $P=0\%$), as it did in the postablation OPTION trial (OR, 0.42 [95% CI, 0.31–0.58]). In the high-bleeding-risk trials, no reduction was observed (OR, 0.93 [95% CI, 0.68–1.27]; $P=79.3\%$). Within this subgroup, PRAGUE-17 favored LAAC (OR, 0.52), whereas CLOSURE-AF favored OAC (OR, 1.16). The common-effect test for subgroup differences was significant ($P=0.0012$), but the random-effects test was not ($P=0.25$), reflecting the heterogeneity introduced

by CLOSURE-AF. Stratification by baseline HAS-BLED score (<3 versus ≥ 3) produced the same pattern. LAAC reduced nonprocedural bleeding compared with warfarin (OR, 0.46 [95% CI, 0.31–0.69], from the PROTECT AF/PREVAIL pooled analysis) and DOACs (OR, 0.61 [95% CI, 0.30–1.24]; $P=83\%$).

Stratification by CHA₂DS₂-VAsC score (≤ 3.5 versus >3.5) and by baseline stroke/transient ischemic attack prevalence (<20% versus $\geq 20\%$) produced identical trial groupings across all 6 RCTs. Point estimates overlapped between subgroups for every stroke outcome (all >0.65).

Subgroup and Interaction Analyses

Subgroup estimates and interaction tests are shown in Table S5 and Figures S23 through S48.

DISCUSSION

The results of our meta-analysis demonstrate that LAAC reduces nonprocedural clinically relevant bleeding in anticoagulation-eligible patients with AF at

Table 3. Pooled Estimates,* Robustness, and Trial Sequential Analysis for LAAC Versus OAC Across 6 Randomized Trials (7028 Patients)

Outcome (k)	Events† (n/N) LAAC vs OAC	OR (95% CI) random effects	95% PI	<i>I</i> ²	LFK	LOO OR range	TSA‡
All stroke (k=6)§	148/3681 vs 111/3323	1.10 (0.78–1.56)	0.60–2.02	10.6%	–1.57	0.96–1.25	29% of RIS adj. 95% CI, 0.66–1.88
Ischemic stroke (k=6)§	126/3681 vs 76/3323	1.41 (1.07–1.86)	0.96–2.08	0%	0.24	1.27–1.49	36% of RIS adj. 95% CI, 0.83–2.41
Hemorrhagic stroke (k=6)§	24/3681 vs 36/3323	0.55 (0.25–1.22)	0.14–2.11	16.4%	–1.67	0.47–0.75	63% of RIS adj. 95% CI, 0.28–1.16
All-cause mortality¶ (k=5)	403/3681 vs 368/3323	0.91 (0.69–1.20)	0.55–1.51	34.7%	–2.59	0.84–1.00	67% of RIS adj. 95% CI, 0.75–1.14
Nonprocedural clinically relevant bleeding¶¶ (k=5)	355/3681 vs 541/3323	0.58 (0.35–0.96)	0.19–1.80	78.8%	0.12	0.50–0.63	78% of RIS adj. 95% CI, 0.48–0.67

*I*² indicates between-study heterogeneity; k, number of trials contributing to the pooled estimate; LAAC, left atrial appendage closure; LFK, Luis Furuya-Kanamori index; LOO OR range, minimum and maximum of the pooled odds ratio estimates across leave-one-out iterations; OAC, oral anticoagulation; OR, odds ratio; PI, prediction interval; RIS, required information size; and TSA, trial sequential analysis.

*All pooled estimates use a random-effects model with Hartung-Knapp adjustment.

†Events are reported as LAAC events/LAAC patients vs OAC events/OAC patients at the longest follow-up available. Forest plots, leave-one-out plots, Doi/LFK plots, and TSA plots for every outcome appear in supplementary documentation.

‡TSA-adjusted CIs account for repeated testing and heterogeneity (*D*² adjustment). The RIS was computed under an O'Brien-Fleming α -spending function, $\alpha = 0.05$, power 80%, and a trial-specific minimally important clinical reduction (mortality 20%, all stroke 20%, ischemic stroke 25%, hemorrhagic stroke 50%, nonprocedural bleeding 35%).

§The 5-year data from PROTECT AF and PREVAIL are reported separately (k=6).

¶PROTECT AF and PREVAIL were pooled at the trial level, yielding 5 contributing units (k=5).

¶¶Analysis combines trial-specific definitions (ISTH major + CRNMB for OPTION and CHAMPION-AF, BARC ≥ 3 for CLOSURE-AF, ISTH major + NMCRB for PRAGUE-17, ICH + transfusion for PROTECT AF + PREVAIL pooled).



standard-to-moderate bleeding risk (eg, those with prior clinically relevant bleeding, a labile or difficult-to-manage anticoagulation course, or an anticipated need for concomitant antiplatelet therapy). This benefit was not observed in the pooled high-bleeding-risk, anticoagulation-intolerant stratum that LAAC has

Table 4. Principal Pooled Subgroup Estimates for LAAC Versus OAC*

Outcome/subgroup	No. of trials	LAAC (n/N)	OAC (n/N)	Random effects OR (95% CI)	<i>I</i> ²
Any stroke (ischemic plus hemorrhagic)					
Overall	6	148/3681	111/3323	1.10 (0.78–1.56)	10.6%
Ischemic stroke					
Overall	6	126/3681	76/3323	1.41 (1.07–1.86)	0%
vs warfarin	2	41/732	14/382	1.56 (0.84–2.90)†	0%
vs DOAC	4	85/2949	62/2941	1.38 (0.91–2.08)	0%
Hemorrhagic stroke					
Overall	6	24/3681	36/3323	0.55 (0.25–1.22)	16.4%
vs warfarin	2	5/732	13/382	0.20 (0.07–0.55)†	0%
vs DOAC	4	19/2949	23/2941	0.82 (0.58–1.16)	0%
All-cause mortality					
Overall	6	403/3681	368/3323	0.91 (0.69–1.20)	34.7%
vs warfarin	2	106/732	73/382	0.72 (0.52–0.99)	0%
vs DOAC	4	297/2949	295/2941	1.00 (0.74–1.35)	2%
Nonprocedural clinically relevant bleeding					
Overall	6	355/3681	541/3323	0.58 (0.35–0.96)	78.8%
Standard bleeding risk	3	202/2231	311/1883	0.53 (0.44–0.64)†	0%
Postablation (OPTION)	1	65/803	137/797	0.42 (0.31–0.58)	...
High bleeding risk	2	88/647	93/643	0.93 (0.68–1.27)†	78.8%

DOAC indicates direct oral anticoagulant; *I*², between-study heterogeneity; LAAC, left atrial appendage closure; OAC, oral anticoagulation; and OR, odds ratio.

*The complete subgroup analysis for every prespecified stratification is provided in Table S5 and Figures S23 through S48.

†Fixed-effect estimate shown for these variables; the random-effects CI was uninformative because of the small number of studies in the subgroup.

traditionally targeted, implying that the current randomized evidence may better support LAAC in patients who could tolerate anticoagulation than in the frail, highest-risk population toward which it is most often directed. This reframing has direct implications for candidate selection, future trial design, and communication of benefits to patients.

Although anticoagulation for stroke prevention in AF is recommended, adherence remains poor in clinical practice: 40% to 50% of eligible patients are untreated or discontinue therapy within 1 to 2 years, driven substantially by hemorrhagic complications.^{23–26} Among patients who do remain on therapy, annual major bleeding rates of 2% to 3% on DOACs and exceeding 3% on warfarin represent a clinically meaningful and modifiable burden.²⁷ This meta-analysis demonstrated a reduction in nonprocedural bleeding by 42% with LAAC compared with OAC; this estimate was robust across leave-one-out iterations, crossed the TSA α -spending boundary, and was driven by studies in anticoagulation-eligible patients at standard bleeding risk and in the postablation population of OPTION, where LAAC was integrated as part of an index ablation procedure.²¹ The CHAMPION-AF trial, which enrolled the largest anticoagulation-eligible population to date, demonstrated similar composite end point efficacy and lower rates of nonprocedural bleeding, reinforcing this conclusion in the patient phenotype where LAAC is most commonly deployed.⁸ The cumulative randomized evidence now provides sufficient support to position LAAC as a legitimate evidence-based strategy to mitigate anticoagulation-related bleeding in certain patients, a conclusion that prior meta-analyses, constrained by smaller sample sizes, could not firmly establish.^{7,28}

Across all 6 randomized trials, however, LAAC was associated with a 41% higher odds of ischemic stroke compared with OAC, a finding notable for its consistency: between-study heterogeneity was zero, every iteration of the leave-one-out OR range favored anticoagulation, and the pattern was unchanged across every prespecified subgroup, including comparator class, device generation, CHA₂DS₂-VASc stratum, and bleeding risk at enrollment. Furthermore, because more than 90% of left atrial thrombi in nonvalvular AF originate within the LAA, any breach in complete mechanical occlusion carries tangible embolic consequences.²⁹ Nonetheless, 2 critical caveats govern interpretation. First, TSA reached only 36% of the RIS without crossing the α -spending boundary, meaning that this signal, however directionally consistent, does not yet meet the threshold for a firm inferential conclusion at a clinically meaningful effect size. Second, the net stroke picture is less alarming than the ischemic stroke figure in isolation: when hemorrhagic and ischemic strokes are combined, the pooled OR for all strokes was 1.10 (95% CI, 0.78–1.56), with no significant difference between arms, reflecting partial hemorrhagic offset. Nevertheless, AF-related ischemic strokes carry a

30-day mortality rate approaching 25% and high rates of permanent disability, making even a statistically inconclusive upward signal clinically important to communicate to patients.³⁰ The practical implication is not that LAAC confers unacceptable stroke risk but that it likely trades hemorrhagic stroke reduction for a modest increase in ischemic events, a trade-off with a net value dependent on the individual patient's baseline hemorrhagic-to-ischemic stroke risk ratio. Shared decision-making anchored to this nuanced trade-off, rather than to aggregated composite end points, is the more intellectually honest framework for patient counseling.

The divergence between CHAMPION-AF and CLOSURE-AF reflects a patient phenotype signal rather than a contradictory result. CHAMPION-AF enrolled anticoagulation-eligible patients with a mean CHA₂DS₂-VASc of 3.5 and mean age of 71.7 years, demonstrating similar major efficacy outcomes and a clear bleeding benefit.⁸ CLOSURE-AF enrolled a distinctly higher-risk population with a mean CHA₂DS₂-VASc score of 5.2, mean age of 77.9 years, and mean HAS-BLED score of 3.1 in whom LAAC did not reduce bleeding (OR, 0.93 [95% CI, 0.68–1.27]).⁹ This finding challenges the prevailing assumption that older, frail, anticoagulation-intolerant patients represent the primary beneficiaries of mechanical occlusion. In this population, ischemic stroke risk is driven substantially by mechanisms beyond the LAA, including aortic arch atheroma, cerebral small-vessel disease, and non-LAA cardioembolism, such that appendage closure provides incomplete protection regardless of technical success.^{31,32} In addition, procedural and post-procedural risks in this group may offset the mechanical benefit of occlusion. The HAS-BLED score, validated for bleeding prediction on anticoagulation, was not designed to identify patients in whom LAAC confers net benefit, and its use as a selection criterion merits reappraisal.³³

Taken together, these findings reframe the relevant clinical question from whether LAAC is noninferior to OAC in stroke prevention to which patients harbor sufficient bleeding burden to derive net benefit from mechanical occlusion while simultaneously carrying a stroke risk profile in which LAA-mediated embolism remains the dominant embolic mechanism. The CHAMPION-AF and CLOSURE-AF populations illustrate that these 2 conditions are not always present together: bleeding burden and LAA-dependent stroke risk do not reliably converge in the oldest and most comorbid patients, where systemic vascular disease and non-LAA embolic mechanisms may dilute the protective signal of appendage closure. A selection framework that integrates stroke mechanism probability alongside bleeding phenotype rather than relying on any single risk score may therefore more precisely identify patients for whom LAAC provides meaningful net benefit.

The hemorrhagic stroke and mortality findings require recontextualization. Across warfarin-controlled trials,

specifically PROTECT AF and PREVAIL, LAAC was associated with substantially lower hemorrhagic stroke and lower all-cause mortality.^{12,18,19} Across DOAC-controlled trials, neither benefit was reproduced, and the comparator-class interaction for hemorrhagic stroke was highly significant ($P=0.0003$). These findings should retire the prior meta-analytic narrative of a broad LAAC mortality and hemorrhagic stroke benefit.^{7,28} The previously published benefit was specific to both device and comparator, generated in an era when warfarin's intracranial hemorrhage burden substantially exceeded what is now achievable with contemporary agents. The 4 large DOAC trials demonstrated a 50% reduction in intracranial hemorrhage relative to warfarin, and in PRAGUE-17, apixaban comprised 95.5% of the anticoagulation arm with correspondingly low hemorrhagic stroke rates.^{20,22,27} CHAMPION-AF and OPTION similarly used physician-directed contemporary DOAC selection.^{8,21} Against this modern comparator, LAAC's mechanical hemorrhagic advantage is substantially attenuated, and future trials and guideline statements must adopt DOACs, not warfarin, as the appropriate reference standard.

The nonprocedural bleeding reduction, although the most statistically mature finding in this analysis, rests on heterogeneous and imperfectly ascertained end points. The pooled estimate combined 3 bleeding definitions (ISTH major plus clinically relevant nonmajor bleeding was used in CHAMPION-AF, OPTION, and PRAGUE-17; Bleeding Academic Research Consortium score ≥ 3 was used in CLOSURE-AF, and intracranial hemorrhage plus transfusion ≥ 2 units was used in the Reddy et al¹² pooled analysis of PROTECT AF and PREVAIL), and all 6 trials carried a Some Concerns rating for the safety end point, reflecting open-label designs and the subjective component of bleeding adjudication. Crossing the RIS on TSA limits the influence of random error but does not correct this systematic ascertainment bias. The bleeding benefit is therefore best characterized as consistent and clinically relevant but not immune to measurement bias.

Several limitations of this analysis deserve explicit acknowledgment. Subgroup analyses rely on trial-level mean CHA₂DS₂-VASC and HAS-BLED scores rather than individual patient data, limiting the precision of risk-stratified estimates; individual patient data pooling was only available for PROTECT AF and PREVAIL via the analysis by Reddy et al¹² and not the 4 contemporary trials. Bleeding end point definitions were heterogeneous across the 5 trials: ISTH major plus clinically relevant nonmajor bleeding (CHAMPION-AF, OPTION, PRAGUE-17), Bleeding Academic Research Consortium score ≥ 3 (CLOSURE-AF), and intracranial hemorrhage plus transfusion ≥ 2 units (the Reddy et al¹² pooled analysis of PROTECT AF and PREVAIL). Sensitivity analysis restricted to the 3 ISTH-defined trials was qualitatively consistent, but residual definition-driven heterogeneity cannot be excluded.³⁴ Device generation and

comparator class are collinear across these data sets: all first-generation Watchman data are derived from warfarin-controlled trials and all Watchman FLX data from DOAC-controlled trials, meaning device effects and comparator effects cannot be cleanly separated. The stratification variables and outcome definitions pooled—CHA₂DS₂-VASC, HAS-BLED, and the trial-specific stroke and bleeding end points—are imperfect and may introduce misclassification. AF suggests but does not prove a cardioembolic mechanism; because aortic arch atheroma, cerebral small-vessel disease, and non-LAA cardioembolism contribute to stroke in this population, appendage closure cannot prevent every event, and its apparent efficacy may be diluted. Furthermore, procedural complications, including pericardial effusion and device embolization, which represent an inherent upfront risk of LAAC not borne by the anticoagulation arm, were not pooled as a primary outcome in this analysis, although they influence the net benefit calculation in shared clinical decision-making. TSA findings for most outcomes (all stroke 29%, ischemic stroke 36%, hemorrhagic stroke 63%, all-cause mortality 67% of the RIS) indicate that the cumulative randomized evidence has not yet reached the threshold for firm inference at a clinically meaningful effect size. Randomized efficacy estimates cannot capture the practical advantages that motivate LAAC: freedom from daily adherence, laboratory monitoring, and drug interactions. Whether these translate into real-world benefit will require pragmatic and registry-based studies. Finally, the prevalence of prior stroke or transient ischemic attack across the included trials ranged from 8% in CHAMPION-AF to 33% in CLOSURE-AF, so the pooled population reflects a mix of primary and secondary prevention, and a prespecified subgrouping by prior-stroke burden showed overlapping estimates. None of these trials was, however, designed for secondary prevention or reported outcomes by prior-stroke status, and dedicated trials in patients with a prior stroke remain warranted.

CONCLUSIONS

This updated systematic review and meta-analysis of 6 RCTs that included 7028 patients demonstrates that LAAC reduces nonprocedural bleeding in anticoagulation-eligible patients with AF, specifically those at standard-to-moderate hemorrhagic risk while receiving OAC. The reduction in nonprocedural bleeding reached its prespecified information size on TSA and is the most statistically mature finding in this analysis; it addresses a major limitation of long-term OAC, namely poor adherence and persistent bleeding risk, while remaining subject to residual bleeding-ascertainment bias. The higher odds of ischemic stroke with LAAC did not reach the RIS on TSA and are therefore inconclusive; overall stroke rates did not differ, reflecting partial offset by lower

hemorrhagic stroke (largely in warfarin-controlled trials). This potential trade-off should be discussed in shared decision-making and addressed by adequately powered trials. LAAC should not be considered a stroke risk reduction substitute for OAC, and its benefit is limited in elderly patients or those at high bleeding risk who have been previously viewed as ideal candidates because studies have shown that neither reduced bleeding risk nor noninferiority has been clearly observed. Prior signals of mortality and hemorrhagic stroke benefit were specific to warfarin-controlled trials and were not reproduced against DOAC comparators, underscoring the importance of using contemporary anticoagulants as the reference standard. Overall, current evidence supports reframing the clinical role of LAAC from a stroke prevention alternative to a strategy primarily reducing non-procedural bleeding in certain patients. This shifts the focus of patient selection toward bleeding phenotype optimization rather than reliance on stroke risk scores. Larger more mechanistically informative trials with longer follow-up are needed before the ischemic stroke, hemorrhagic stroke, and mortality comparisons can be considered settled and to clarify the net clinical benefit of LAAC across patient subgroups.

ARTICLE INFORMATION

Received June 16, 2026; final revision received August 21, 2026; accepted September 9, 2026.

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Acknowledgments

The authors thank Kristin Kraus and Cortlynd Olsen for editorial assistance.

Sources of Funding

None.

Disclosures

A. Shoskes reports compensation from the American Stroke Association and grants from the National Institutes of Health (NIH). Dr Vijayaraman has received honoraria and research support from Medtronic; serves as a consultant and advisory board member for Medtronic and Abbott; has received honoraria from Biontronik and Boston Scientific; and holds a patent for a His bundle pacing delivery tool. Dr Majersik has received institutional grants from NIH/National Institute of Neurological Disorders and Stroke, American Heart Association (AHA), and NIH/National Center for Advancing Translational Sciences and honoraria from the American Academy of Physical Medicine and Rehabilitation; serves on the UC Davis CTSA External Advisory Board, UT Houston CTSA External Advisory

Board, and U Mass CTSA External Advisory Board and the BrainPAC Executive Committee; is a consultant for UpToDate; and is an Associate Editor of the AHA journal *Stroke*. Dr Tandar reports compensation from Edwards Lifesciences for consultant services and grants from W. L. Gore & Associates, Inc Ramesh Grandhi is a consultant for Balt USA, LLC; Cerenovus; Kaneka; Integra LifeSciences Corporation; Medtronic Neurovascular; Rapid Medical, Ltd.; and Stryker Neurovascular and received an institutional educational grant from Scientia Neurovascular. The other authors declare no conflicts of interest.

Supplemental Material

Supplemental Methods
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